Dev S Panchal

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Summary

Applied AI / GenAI Engineer specializing in LLMs, RAG pipelines, and agentic workflows, shipping Generative AI systems end-to-end from data pipelines to production. Proficient in Python, with hands-on experience across LangChain and vector databases. Thrives in fast-moving, ambiguous environments, driven by ownership over outcomes, not just code. Curiosity-driven, always digging past "it runs" to "why it runs.

Experience

AI Engineer *Ahmedabad, India*
Relig Global (Hybrid) *Nov. 2025 - Present*

- Designed and deployed Python-based ML pipelines for financial data, transforming raw, unstructured datasets into clean, structured signals used directly in trading and operational decision-making.
- Built a retrieval pipeline over unstructured financial reports and client documents, applying RAG techniques to convert raw PDFs into structured, queryable context for downstream use.
- Automated data reconciliation and reporting workflows using Python (openpyxl), cutting weekly turnaround from 3 days to same-day delivery, a 66% reduction in processing time.
- Partnered directly with stakeholders to convert ambiguous business requirements into shipped ML and GenAI-driven automation solutions, owning problems end-to-end across engineering and product.

Education

LJ University *July 2023 - Present*
B.Eng in Artificial Intelligence and Data Science, CGPA: 8.95/10

Projects

ScholarBot - Intelligent School FAQ Assistant *Jan 2026*

- Architected a hybrid conversational AI system combining a rule-based intent engine (tokenization, normalization, confidence scoring) with a Gemini 2.5-flash fallback; integrated a PDF ingestion pipeline using text extraction and sliding-window chunking to parse school documents (timetables, notices, rulebooks) into retrievable context for document-grounded LLM responses.
- Built a Flask REST backend with a layered architecture of NLP preprocessing, PDF parsing, intent classification, and LLM orchestration as discrete modules and also implementing confidence-threshold routing so Gemini API is invoked only on ambiguous or out-of-distribution queries, reducing unnecessary API calls.

PredictGrad - Academic Risk Detection with ML *Jun 2025*
thepredictgrad.streamlit.app

- Developed end-to-end regression and classification workflows using ensemble methods (Ridge, CatBoost, ExtraTrees, BalancedBagging with Voting/Stacking) to predict scores and identify students at academic risk.
- Built an interpretable risk analysis dashboard powered by SHAP insights; attained MAE between 5.16–7.10 and an F1-score of 0.51.

Technologies

AI/GenAI: LLMs, Gemini API, RAG, LangChain, Vector Databases (Chroma), Prompt Engineering
Core: Python, Django, FastAPI, REST APIs
ML/DL: Scikit-learn, TensorFlow, Keras
Data & Infra: PostgreSQL, MySQL, MongoDB, Git, GitLab, Docker

Skills

RAG Pipelines, Agentic AI, Generative AI, Vector Databases, Prompt Engineering, OpenAI API, Gemini API, Machine Learning, Deep Learning, Data Pipelines & ETL, Django Backend Development, Full-Stack Prototyping, SQL, NoSQL